Fuel 246 (2019) 319–329

Contents lists available at ScienceDirect

Fuel

journal homepage: www.elsevier.com/locate/fuel

Synergistic effects of co-pyrolysis of macroalgae and polyvinyl chloride on bio-oil/bio-char properties and transferring regularity of chlorine

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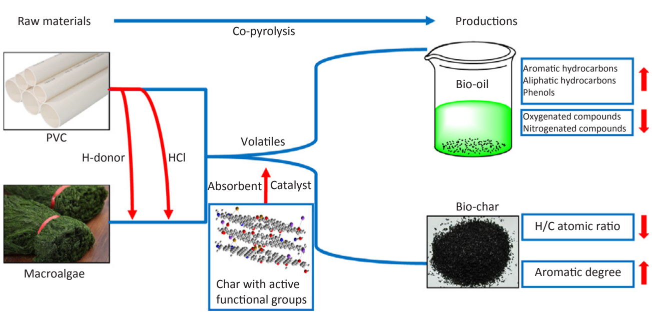
GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Co-pyrolysis
Macroalgae
Polyvinyl chloride
GC–MS
Chlorine

ABSTRACT

The present study aimed to investigate synergistic effects on properties of bio-oil/bio-char and transferring regularity of chlorine during co-pyrolysis of macroalgae *Enteromorpha clathrata* (EN) mixed with different mass percentage of polyvinyl chloride (PVC) (5 wt%, 10 wt%, 15 wt%, 20 wt%, and 25 wt%) in a fixed bed reactor at 550 °C. The results showed that the experimental values of bio-oils were 0.43–2.57 wt% lower than the calculated values, while the bio-chars were 1.28–4.93 wt% higher. From the GC–MS analysis of bio-oils, the relative contents of oxygenated and nitrogenated compounds were reduced significantly while those of phenols, aromatic and aliphatic compounds were increased dramatically during co-pyrolysis of EN and PVC. Interaction during co-pyrolysis of EN and PVC decreased the H/C atomic ratio of bio-chars, enhancing the aromatic degree. In addition, the chlorine distribution in bio-chars increased whereas decreased in bio-oils and non-condensable gas with the addition content of PVC increasing. The results showed that a majority of chlorine from EN-PVC blends was trapped in bio-char products, which was favorable to the utilization of bio-oils and friendly to pyrolysis equipment and environment.

Abbreviations: EN, *Enteromorpha clathrata*; PVC, polyvinyl chloride; Exp., the experimental value; Cal., the calculated value

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https://doi.org/10.1016/j.fuel.2019.02.037
Received 19 November 2018; Received in revised form 14 January 2019; Accepted 7 February 2019
Available online 04 March 2019
0016-2361/

1. Introduction

The utilization of algae, as one of the most potential sources of third-generation bio-fuel feedstock [1], has been widely investigated due to its fast growth rate and good reproduction ability [2,3]. Macroalgae can grow up to 60 m in length, which is a considerable size. The average photosynthetic efficiency of algae biomass is much higher than that of terrestrial biomass. In addition, annual primary production rates for the macroalgae are higher than those for terrestrial biomass [4]. Therefore, there could be algae blooms without right processing methods, which leads to water pollution.

At present, plastic found to be one major pollutant to the environment which are produced humungous for manufacturing bags, containers and various construction materials [5]. The used plastics could generate a large amount of municipal solid waste (MSW). The different kinds of plastics include low density polyethylene (LDPE), high density polyethylene (HDPE), polypropylene (PP), polyvinyl chloride (PVC), polystyrene (PS) and polyethylene terephthalate (PET), which altogether account for 74.2 wt% of the European plastic demand in 2012 [5,6]. Plastic waste disposal through direct combustion is not encouraged due to the formation of toxic pollutants such as SO_2, NOx, heavy metals and particulates in both flue gas and solid residues. Thus, it becomes a big challenge how to deal with the large amount of MSW in an environmentally-friendly and pollution free manner. In addition, Cl-containing plastic waste could generate severe pollution by releasing chlorinated compounds and dioxins. This could lead to troublesome process complications such as corrosion of equipment and catalyst poisoning [7].

At present, pyrolysis is regarded as the most promising thermal approach for plastic waste and algal/terrestrial biomass utilization [8–12]. Wang S et al. has investigated the pyrolysis process of two kinds of seaweeds (*Enteromorpha clathrate* and *Sargassum natans*) and analyzed the chemical components of produced bio-oils [13]. The fast pyrolysis characteristics of *Saccharina japonica* were investigated using a thermogravimetric analyzer and a bubbling fluidized-bed reactor by Choi JH et al. [8]. Zhou J et al. investigated the pyrolysis mechanism of polyvinyl chloride (PVC) by detecting the characteristics of chars [14]. López A et al. have investigated the pyrolysis characteristics of a mixture of polyethylene (PE), polypropylene (PP), polystyrene (PS), polyethylene terephthalate (PET) and polyvinyl chloride (PVC) [15]. However, the pyrolysis oil obtained from algal or woody biomass has undesired properties in terms of bio-fuel quality, such as high content of oxygenated and nitrogenated compounds, low heating values, rich in organic acids.

Plastic materials which are rich in carbon and hydrogen could act as a hydrogen donor to the biomass in the pyrolysis process [16]. Studies have shown that co-pyrolysis of woody biomass and plastics without Cl-containing compounds could increase pyrolysis oil quantity and quality compared to pyrolysis of separate components [10,17,18]. Although there are many literatures studying the co-pyrolysis of woody biomass and plastics without Cl-containing compounds, very few studies could be found on the co-pyrolysis of algal biomass and plastics without Cl-containing compounds.

Wu X et al. have investigated co-pyrolysis kinetics and behavior of microalgae and polypropylene (PP) via TG-FTIR and TG-MS. The results showed that PP could accelerate the pyrolysis process of microalgae and a significant decrease in CO_2 occurred during the co-pyrolysis of the two kinds of materials [19]. Azizi K et al. have also investigated the thermal decomposition behavior and kinetics of microalgae *Chlorella vulgaris* and polypropylene (PP). TGA analysis showed that there was a synergistic effect when microalgae and PP were pyrolyzed together [20]. In addition, there are less literatures on the co-pyrolysis of macroalgae and plastics. Exploring the literatures, only Kositkanawuth K et al. investigated the co-pyrolysis behavior of macroalgae *Sargassum* and polystyrene (PS) [21].

Although pyrolysis of PVC is limited due to the high capital costs, hydrogen chloride (HCl) released from PVC might function as acid-catalyst affecting the pyrolysis process of biomass. In addition, PVC might act as H-donors and lead to hydrogen transferring from its chains to biomass derived radicals [22]. The HCl and H* could accelerate the interaction in the co-pyrolysis of PVC and biomass, while co-pyrolysis of biomass and PVC has rarely been studied.

Tang Y et al. have investigated the co-pyrolysis characteristics and kinetic analysis of typical organic food waste soybean protein and polyvinyl chloride (PVC) [7]. TGA analysis showed that the activation energies needed in co-pyrolysis were decreased by 2–13% due to the interaction of soybean protein and PVC. The fixed-bed experiments indicated that interactions between the two kinds of materials could reduce the yields of tar and promote the yields of char compared with linear calculation. Desirably, nitrogen-containing compounds in tars obtained from co-pyrolysis of soybean protein and PVC were reduced by 61–93%. Dai M et al. have investigated the co-pyrolysis behaviors and product characteristics of microalgae *Chlorella vulgaris* and polyvinyl chloride [23]. There was a remarkable synergistic effect between microalgae and PVC during co-pyrolysis in terms of quantity and quality of bio-oils and bio-chars. The contents of nitrogen-containing compounds and oxygenated organic matters in bio-oils reduced significantly and the heating values of bio-chars increased. The above literatures indicated that synergistic effects were existed during co-pyrolysis of protein/microalgae and PVC, which could promote the quality of bio-oils and bio-chars. However, to the best of our knowledge, little studies of co-pyrolysis of macroalgae and PVC have been conducted. The chlorine distribution in bio-oils, bio-chars and non-condensable gas during co-pyrolysis of macroalgae and PVC was not clear, and the quality and chemical compounds of the three phase products were also not investigated before by other researchers. Therefore, the present research was aimed to fill in those gaps.

This study aimed to investigate the synergistic effects on properties of bio-oils and bio-chars, and transferring regularity of chlorine during co-pyrolysis of macroalgae *Enteromorpha clathrata* and polyvinyl chloride. The addition of PVC accounted for 5 wt%, 10 wt%, 15 wt%, 20 wt%, and 25 wt% of the feedstock mixtures. Thermogravimetric analysis was firstly employed to investigate the thermal decomposition behavior of EN and PVC. Then the pyrolysis experiments of EN-PVC blends were performed in a fixed bed reactor at the temperature of 550 °C. The chemical compounds of bio-oils were analyzed through GC–MS analysis. In order to determine the synergistic effects systematically, the elemental analysis and heating values of bio-chars were also measured. The contents of chlorine in bio-oils and bio-chars were measured via potentiometric titration. The experimental results were compared with the linear calculated values to investigate the synergistic effects.

2. Materials and methods

2.1. Materials

Enteromorpha clathrata was collected from Zhanjiang coast of South China Sea, Guangdong Province, China. The samples were dried under the sun for a week and pulverized to a 0.3 mm particle size. Then, they were stored in beakers and dried in oven at 105 °C for 24 h. PVC powder was purchased from Wenzhou Online Chemical Store, China. The particle size of PVC was below 0.3 mm. Before pyrolysis experiments, PVC was also dried in oven at 105 °C for 24 h. The formula of pure PVC is $(C_2H_3Cl)_n$, therefore the element contents of C, H, Cl by calculation are 38.4%, 4.8% and 56.8%, respectively. Different mass percentages of PVC (5 wt%, 10 wt%, 15 wt%, 20 wt%, and 25 wt%) were mixed with EN to study the co-pyrolysis behaviors.

Proximate and ultimate analyses of EN and PVC were carried out using elemental analyzer according to the standards GB212-91, GB476-91, and GB/T213-2003. The results are shown in Table 1. The content of Cl element in PVC was detected through potentiometric titration

Table 1
Proximate, ultimate and heating value analyses of biomass and plastic used in this study.

	EN	PVC
Proximate analysis (as-received basis)		
Ash content (wt%)	25.24	0
Volatile content (wt%)	53.45	96.41
Moisture content (wt%)	12.91	0.17
Fixed carbon (wt%)	8.40	3.42
$Q_{net,\ ar}$ (MJ/kg)	15.47	21.66
Ultimate analysis (dry ash-free basis)		
C (wt%)	32.71	38.19
H (wt%)	5.38	4.94
O (wt%)	51.85[a]	–
N (wt%)	4.85	–
S (wt%)	2.01	–
Cl (wt%)	3.20	47.66

[a] By difference.

according to GB/T 7139-2002. In addition, the functional groups in EN and PVC were determined by Fourier transform infrared spectroscopy (FTIR, Nicolet Nexus470 1.2).

2.2. Thermogravimetric analysis

The thermogravimetric analyzer used in the experiments was STA449C instrument from NETZSCH-Gerätebau GmbH. In a typical analysis, 5–10 mg of samples was placed into an alumina crucible and pyrolyzed in a high purity nitrogen gas at a flow rate of 100 ml/min. Temperature was raised from room temperature to 1000 °C at a constant heating rate of 20 °C·min^{-1}.

2.3. Pyrolysis process

The pyrolysis experiments were performed in a fixed bed stainless reactor. The schematic diagram was shown in Fig. 1. Nitrogen gas was used as the carrier gas. As soon as the temperature of the N_2 preheating equipment was up to 550 °C, N_2 was fed at a flow rate of 0.8 NL·min^{-1} to remove air in the reactor before the pyrolysis of feedstock. After the reactor was heated to 550 °C, the prepared algae sample or the mixture of algae and PVC samples, stored in the sample storage connected with the fixed bed reactor, was fed into the reactor after the N_2 flow rate was adjusted to 0.2 NL·min^{-1}. The condensable volatiles were collected in the condenser as bio-oil. Following the condenser, 0.5 mol/L NaOH solution was used to absorb HCl gas released from the co-pyrolysis process. Then, the non-condensable gas without HCl was flowed into a gas bag. The solid residues were collected in the fixed bed reactor as the bio-char. The weight of bio-oil and bio-char was directly measured. The amount of gas products was calculated by subtracting the bio-oil and bio-char yields from the weight of initial algae or the mixture of algae and PVC sample.

The yields of solid, liquid and gaseous products from the co-pyrolysis in the experiments were then calculated using Eqs. (1)–(3) below:

$$\text{Bio-oil yield (wt. \%)} = \frac{W_{liquid}}{W_{tol.}} \times 100\% \tag{1}$$

$$\text{Bio-char yield (wt. \%)} = \frac{W_{solid}}{W_{tol.}} \times 100\% \tag{2}$$

$$\text{Non-condensable gas yield (wt.\%)} = 100\% - \text{Bio-oil yield (wt.\%)} - \text{Bio-char yield (wt.\%)} \tag{3}$$

where W_{liquid} and W_{solid} represent weight of bio-oil and bio-char (g), respectively and $W_{tol.}$ represents the dry weight of injected biomass (g).

The synergistic effect of co-pyrolysis was explored by comparing the linear calculated value and the experimental. The linear calculated values of each product (T_0) were according to the following Eq. (4);

$$T_0 = W_1T_1 + W_2T_2 \tag{4}$$

where, T_1 and T_2 are the product yields from individual pyrolysis of EN and PVC, respectively, while w_1 and w_2 are the mass percentages of EN and PVC in an experiment run.

All pyrolysis experiments were performed in triplicate and the differences between each experimental yield values were lower than 5%.

2.4. Chemical characteristics of bio-oil and bio-char

The algae bio-oils were analyzed through a gas chromatograph (Agilent Technologies, 7890A) coupled with a mass spectrometer (Agilent Technologies, 5975C) with a triple-axis detector. The GC system was equipped with an Agilent HP-5 UI column. The size of the column is: 30 m (length) × 250 μm (diameter) × 0.25 μm (thickness). High-purity helium was used as the carrier gas at flow rate of 3 mL·min^{-1}. After the temperature of GC injector was set at 300 °C, the mixture containing of pyrolysis oils and anhydrous alcohol was injected into the GC–MS system of injection volume 1 μL. The heating procedure

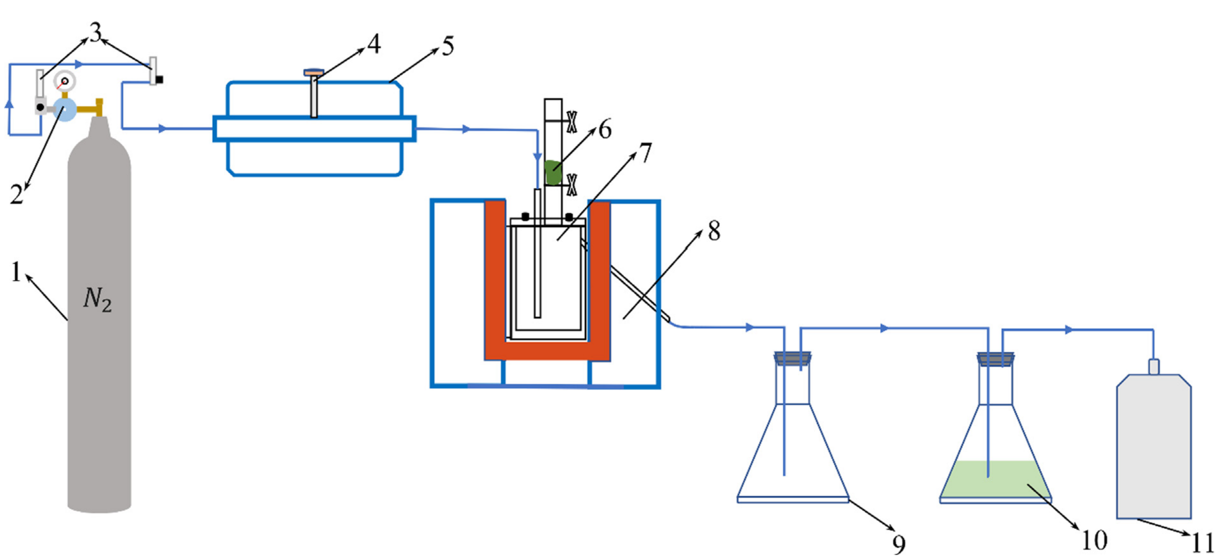

Fig. 1. Schematic diagram of a fixed bed pyrolysis apparatus. 1. Nitrogen cylinder; 2. Gas-pressure gauge; 3. Flowmeter; 4. Thermocouple; 5. N_2 preheating equipment; 6. Sample; 7. Fixed bed reactor; 8. Furnace; 9. Bio-oil storage; 10. 0.5 mol/L NaOH solution; 11. Gas bag.

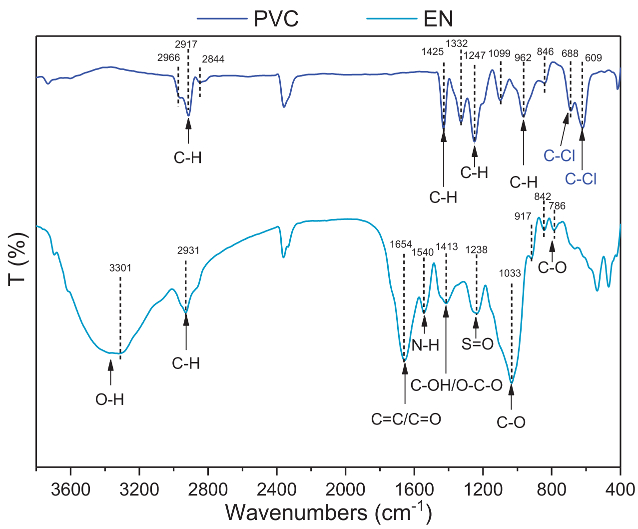

Fig.2. FTIR spectra of EN and PVC.

of the GC oven was pre-programmed from initial temperature of 40 °C to 300 °C at a ramping heating rate of 10 °C·min^{-1}. It was maintained at 300 °C for 5 min.

In addition, the elemental composition and high heating values of bio-chars obtained from co-pyrolysis of EN and PVC were measured according to the standards GB476-91 and GB/T213-2003.

The experimental H/C atomic ratio was estimated according to Eq. (5);

$$H/C_{ratio,Exp.} = \frac{12\omega(H)}{\omega(C)} \tag{5}$$

where $\omega(H)$ and $\omega(C)$ mean the elemental contents in bio-char (wt%).

The calculated H/C atomic ratio was determined according to Eq. (6) [22];

$$H/C_{ratio,Cal.} = H/C_{EN,ratio,Exp.}W_1 + H/C_{PVC,ratio,Exp.}W_2 \tag{6}$$

2.5. Cl content in pyrolysis products

The Cl contents in bio-oil and bio-char products were determined by potentiometric titration. The instrument is ZD-3A automatic potentiometric titrator from Shanghai Anting electronic instrument factory, China. The Cl content in non-condensable gas was calculated by difference.

The organic Cl in bio-oils or bio-chars should be converted into Cl^{-1} existing in solutions through ionization firstly. The performance steps were summarily illustrated as follows; Firstly, the weighed object (bio-oil or bio-char) was placed on a filter paper followed by marking the folding line and clamped on the platinum helix after folding it. Then, water (20 ml), sodium hydroxide solution (1 ml) and hydrogen peroxide solution (0.15 ml) were added to the flask and make oxygen flow to the flask for 5 min. The bottle plug with burning filter paper was infixed quickly into the combustion bottle after igniting the filter paper and flask was kept upside down while burning. Then the flask was gently shaken under the cold water. After 30 min, the flask was opened. Rinsed and the final volume was made to 60 ml followed by the addition of sodium nitrate (1 g) and nitric acid solution (2.5 ml). The above steps were repeated without adding sample to make blank control. In the above operations, inorganic Cl was also dissolved in the solution. Finally, the solutions with Cl^{-} were determined on ZD-3A automatic potentiometric titrator. The mass of Cl was measured according to Eq. (7);

$$X_i = \frac{c(V_1 - V_2)M}{m_i} \times 100\% \tag{7}$$

where X_i represents the mass percentages of Cl in bio-oil, bio-char (wt %), V_1 and V_2 represent the volume of $AgNO_3$ standard solution used in the non-control and control experiments, respectively (mL). c is the concentration of $AgNO_3$ standard solution (0.05 mol/L). m_i is the mass of samples for analyses (mg). M means the molar mass of Cl^{-1} (35.453 g/mol).

The experimental Cl distribution in three phase products was calculated based on Eq. (8);

$$F_i = \frac{X_i W_i}{m_{tol.Cl}} \tag{8}$$

where F_i represents the experimental Cl distribution in three phases (wt %). W_i represents the weights of bio-oils or bio-chars (mg), and $m_{tol.Cl}$ depicts the total mass weight of Cl in EN-PVC blends in corresponding experimental runs (mg).

The calculated Cl distribution in three phase products was calculated based on Eq. (9);

$$F_i' = \frac{w_1 X_{i,EN} W_{i,EN} + w_2 X_{i,PVC} W_{i,PVC}}{m'_{tol.Cl}} \tag{9}$$

where F_i' represents the calculated Cl distribution in three phases (wt %). w_1 and w_2 are the mass percentages of EN and PVC in each run. $X_{i,EN}$ means the mass percentages of Cl in EN bio-oil and bio-char (wt %), and $X_{i,PVC}$ is the mass percentages of Cl in PVC bio-oil and bio-char (wt%). $W_{i,EN}$ means the weights of EN bio-oil or bio-char, while $W_{i,PVC}$ means the weights of PVC bio-oil or bio-char (mg). $m'_{tol.Cl}$ means the total mass weight of Cl in EN-PVC blends in corresponding experimental runs through calculation (mg).

3. Results and discussions

3.1. Thermogravimetric analysis of EN and PVC

3.1.1. FTIR characteristics of EN and PVC

Raw materials (EN and PVC) were analyzed by FTIR to gain a clear understanding on their functional groups. The FTIR spectra are shown in Fig. 2 and the peak assignments in FTIR spectra were summarized in Table 2. For the spectra of PVC, the special absorption bands at 688 and 609 cm^{-1} were assigned to C–Cl stretching from group –CHCl–. The main bio-chemical components of EN are polysaccharides, proteins, and lipids. The functional groups in those components were determined from FTIR spectra of EN. The absorption band at 3301 cm^{-1} was ascribed to O–H stretching vibration at polysaccharides and lipids. The

Table 2
Summary of peak assignments in FTIR spectra of EN and PVC.

Wavenumbers (cm^{-1})	Assignment	Refs.
3301	–OH, O–H stretching	[2]
2966	–CHCl–, C–H stretching	[14,26–28]
2931, 2917	–CH$_2$–, C–H asymmetric stretching	[2,14,26–28]
2844	–CH$_2$–, C–H symmetric stretching	[14,26–28]
1654	C=C/C=O stretching	[2]
1540	N–H stretching in protein	[2]
1425	–CH$_2$–, C–H bending	[14,26–28]
1413	C–OH stretching and O–C–O symmetric stretching	[2,25]
1332	–CHCl–, C–H bending	[14,26–28]
1247	–CHCl–, C–H bending	[14,26–28]
1238	S=O stretching	[2,29]
1099	Skeletal vibration	[30,31]
1033	C–O stretching	[2,25]
962	–CH$_2$–, C–H rocking	[14,26–28]
917	C–O stretching	[2,24]
846	–CH$_2$–, C–H rocking	[14,26–28]
842	C–O stretching	[2,24]
786	C–O stretching	[2,24]
688	–CHCl–, C–Cl stretching, Cl trans to H	[14,26–28]
609	–CHCl–, C–Cl stretching, Cl trans to C	[14,26–28]

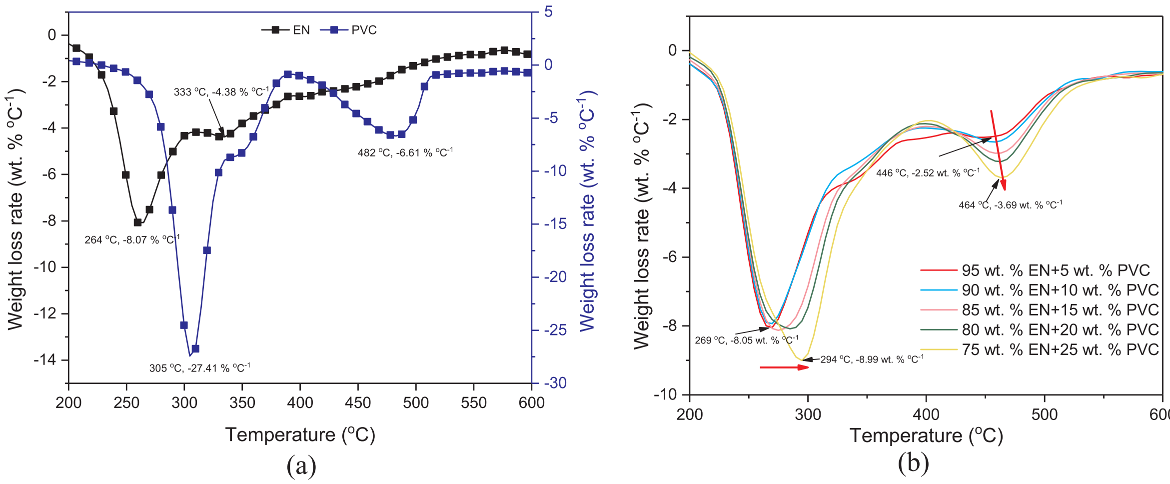

Fig. 3. DTG curves of (a) individual feedstocks and (b) EN-PVC blends.

bands at 1654 and 1540 cm^{-1} indicate the existence of protein in EN [2]. The absorption area between 950 and 750 cm^{-1} was the characteristic of carbohydrates [24], such as the absorption band at 917, 842, 788 cm^{-1} associated with C–O stretching vibrations in uronic acid and mannuronic acid [25].

3.1.2. Thermal decomposition behavior of EN and PVC

Differential thermogravimetric (DTG) curves of individual feedstocks and mixtures of EN with different mass ratios of PVC (5 wt%, 10 wt%, 15 wt%, 20 wt%, and 25 wt%) are shown in Fig. 3. From DTG curves of EN and PVC, it was observed that their thermal decomposition behaviors were different from each other. The devolatilization of EN and PVC mainly occurred between 200 and 600 °C.

In Fig. 3(a), the maximum weight loss rate of EN was −8.08 wt% $°C^{-1}$ at 264 °C. This peak could be attributed to the thermal decomposition of soluble polysaccharides, amino acids, fatty acids and proteins in EN. A shoulder at 333 °C could be attributed to the presence of most resistant thermal components such as cellulose in EN structure [32]. A series of complex reactions took place, including bond scission, dehydration, free radical formation, creation of oxygenated compounds and char formation. The decomposition behavior was mainly observed before 550 °C.

The DTG curves of PVC showed two distinct peaks at 305 °C (−27.41 wt% $°C^{-1}$) and 482 °C (−6.61 wt% $°C^{-1}$) in the temperature range between 235 °C and 520 °C. The main weight loss of PVC was between 230 and 390 °C. In this stage, the main reaction was the dehydrochlorination of PVC polymer forming de-HCl PVC and volatiles [22,33]. The volatiles mainly consist of HCl, small amounts of benzene, toluene and other hydrocarbons [5]. The second stage at the temperature range of 390–520 °C corresponded to the cracking and decomposition of the de-HCl PVC and the formation of chars.

It was seen from Fig. 3(b) that two main stages occurred in the thermal decomposition of the EN-PVC blends. The first stage (200–400 °C) was attributed to the dehydrochlorination of PVC and the decomposition of carbohydrates and proteins in EN. In this stage, the compounds released from carbohydrates and proteins could interact with HCl and other light molecules from PVC. The second stage (400–550 °C) mainly associated with the decomposition of the de-HCl PVC. The first peak shifted to a higher temperature range when the addition of PVC increased from 5 to 25 wt%, and the maximum weight loss rates of the second peaks increased with the increasing addition of PVC. The results indicated that mixing EN and PVC had changed the pyrolysis reaction, which might be attributed to the release of HCl from PVC, as an acid catalyst, and the EN bio-chars with large amounts of metal oxides and surface chemical compounds. In order to have a further investigation on the synergistic effects, co-pyrolysis experiments in a fixed bed reactor were carried out at 550 °C to determine the products yields and characteristics of bio-oils and bio-chars.

3.2. Pyrolysis experiments

Individual pyrolysis and co-pyrolysis of EN and PVC were performed in a fixed bed reactor at the temperature of 550 °C. The yields of products, bio-oils, bio-chars and non-condensable gas, were illustrated in Fig. 4. The yields of bio-oil, bio-char and non-condensable gas in the pyrolysis of EN were 44.78 wt%, 41.71 wt%, and 13.51 wt%, while those in the pyrolysis of PVC were 28.35 wt%, 5.74 wt%, and 65.91 wt%, respectively. As the addition of PVC increased from 5 wt% to 25 wt%, the yields of bio-oil and bio-char decreased by 1.25–6.58 wt% and 0.52–4.06 wt% while that of non-condensable gas increased by 1.77–10.64 wt%. The reason for this phenomenon was mainly due to the material characteristics of PVC. From Table 1, the volatile content of PVC is high up to 96.41 wt%, while the fixed carbon is 3.42 wt% and there is no ash content. This could result in the more gas phase product but less solid phase product in the pyrolysis process of PVC. Therefore, the yields of bio-char and bio-oil decreased while gas increased in the co-pyrolysis of EN and PVC.

To further investigate the synergistic effects on the yields of bio-oils and bio-chars, the experimental values were compared with the results obtained by linear calculation, as shown in Fig. 5. The experimental values of bio-oils were lower 0.43–2.57 wt% than the calculated values,

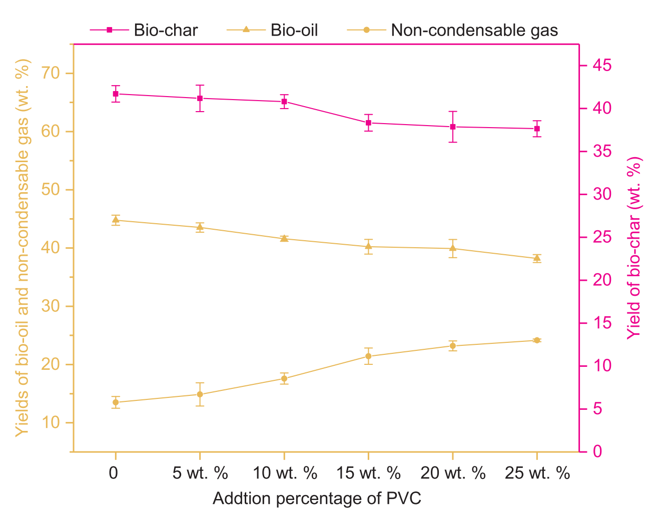

Fig. 4. Product yields from the co-pyrolysis of EN and PVC with the addition contents of PVC.

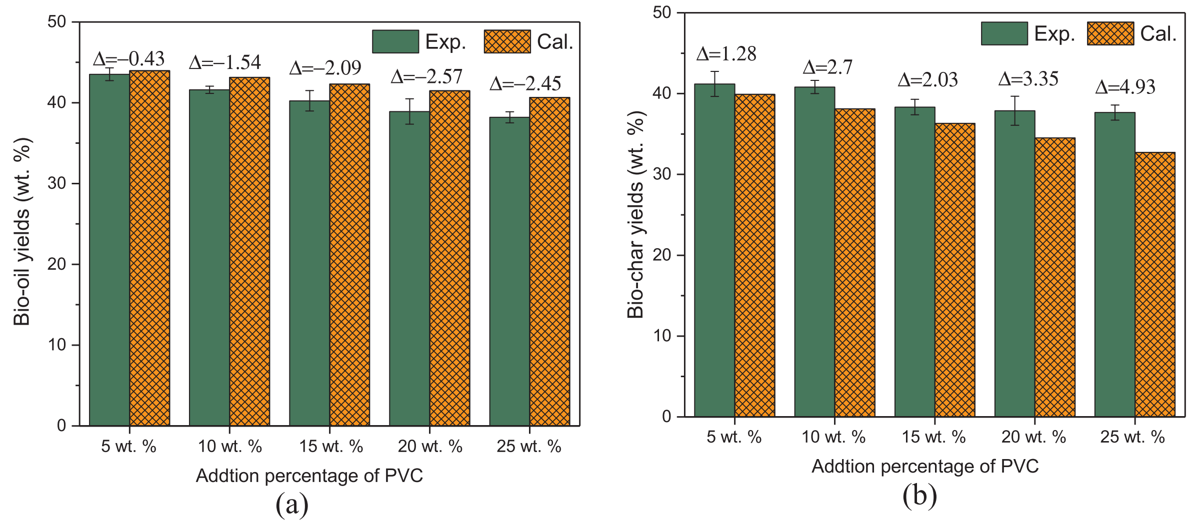

Fig. 5. Comparison of the experimental and calculated values of yields of bio-oils (a) and bio-chars (b). (Δ = Exp.−Cal.)

while the experimental values of bio-chars were higher 1.28–4.93 wt% than the calculated values. This phenomenon indicated that co-pyrolysis of EN and PVC would reduce the yields of bio-oils and increased the yields of bio-chars. This is similar with results from other previous researches by Dai M et al. [23], who revealed that the co-pyrolysis of microalgae and PVC yielded more bio-chars but less bio-oils than the expected values. In addition, as the content of PVC increased from 5 wt % to 25 wt%, the interaction was even more obvious since the difference between the experimental and calculated values increased for both bio-oils and bio-chars. This phenomenon was also observed by Tang Y et al. [7] and Dai M et al. [23]. The reasons for the yield differences between the experimental and calculated values are mainly as following; The HCl obtained from PVC possibly functioned as acid-catalyst to suppress depolymerization and accelerate dehydration and charring progress during co-pyrolysis process [7].

3.3. Characterization of bio-oil from co-pyrolysis

The chemical compositions of bio-oils from individual pyrolysis and co-pyrolysis of EN and PVC were detected by GC–MS analysis, which could provide the structural information at a molecular level. The chromatograms of bio-oils from individual pyrolysis of EN and PVC were illustrated in Fig. 6. The components of EN bio-oil were mainly alcohols, ketones, phenols, organic acids, nitrogen-containing compounds and other oxygenated compounds (furfural and 5-methyl-2-furancarboxaldehyde) while the main components of PVC oil were the aromatic hydrocarbons, including heterocyclic aromatics, light aromatics (1 ring) and light polycyclic aromatic hydrocarbons (PAHs) compounds (2–3 rings). EN contains three main components, namely, carbohydrates, proteins and lipids. The dehydration of carbohydrates could produce furan derivates and anhydrosugars in EN bio-oils. N-containing compounds, such as, amines and amides, nitriles, and pyridines, were from the cracking of proteins, decomposition and condensation of amino acids and dehydration, cyclization and Maillard reaction of N-containing reactive intermediates. The cracking and steam reforming of lipids produced the aliphatic hydrocarbons, long-chain carboxylic acids and long-chain esters [2,13]. There existed isomerization and aromatization reactions via several steps for PVC pyrolysis volatiles to form the aromatic and aliphatic compounds, in which aromatized reaction was the dominant role. The light aromatic hydrocarbons were mainly toluene and p-xylene, and the light polycyclic aromatic hydrocarbons were mainly naphthalene, phenanthrene, anthracene and their derivatives. There was no oxygenated and nitrogenated compounds in PVC bio-oil due to its elemental characteristics.

The chemical groups of bio-oils obtained from co-pyrolysis of EN and PVC were divided into five groups: aromatic hydrocarbons, aliphatic hydrocarbons, phenols, other oxygenated compounds and

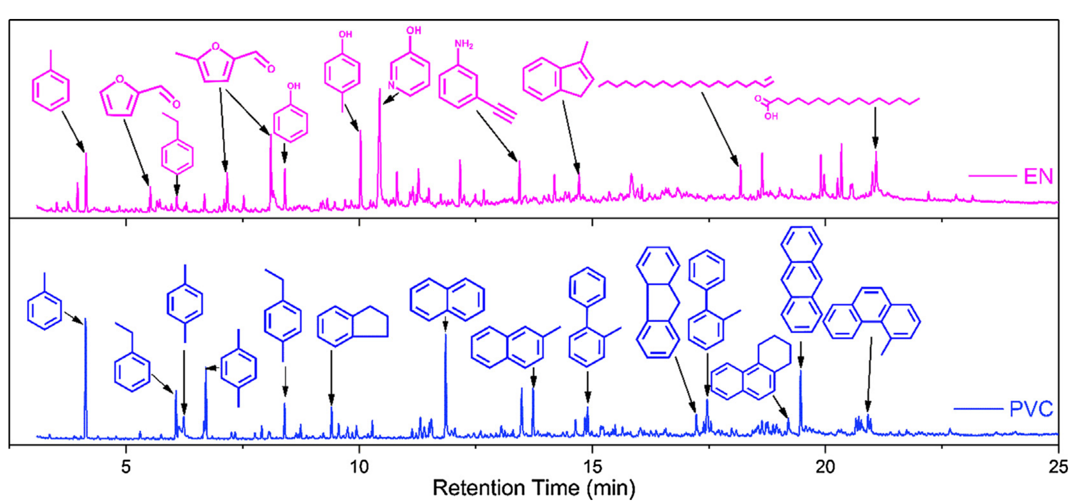

Fig. 6. GC–MS analysis of bio-oils obtained from individual pyrolysis of EN and PVC.

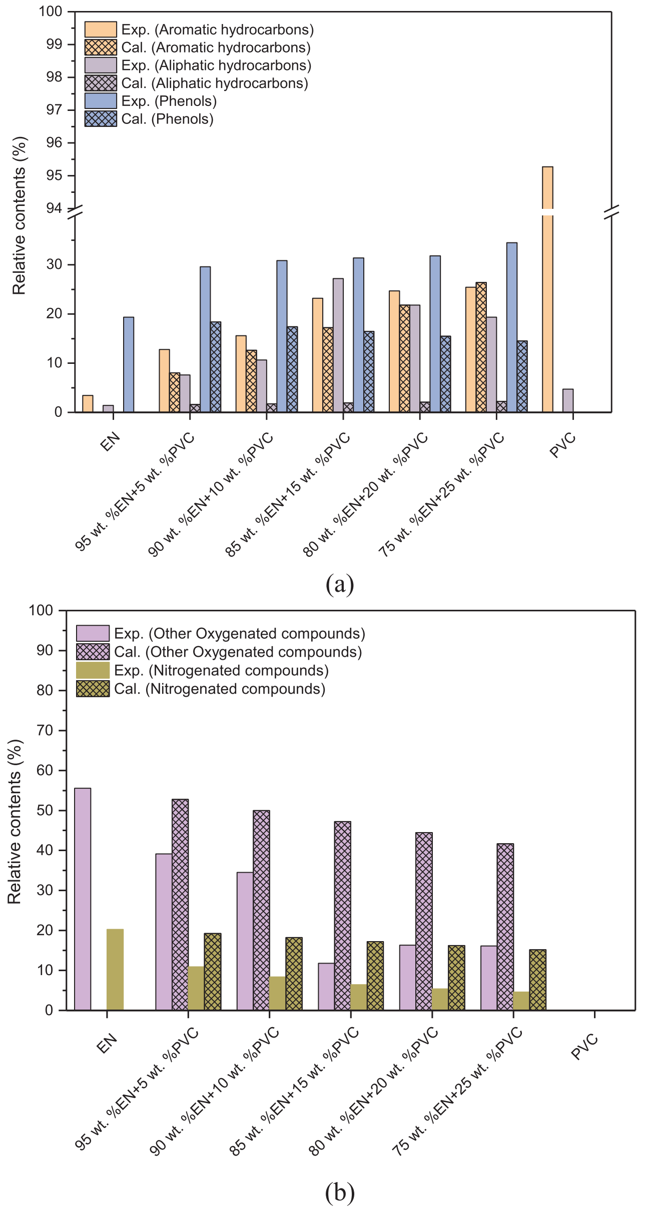

Fig. 7. Relative contents of different groups of the bio-oils from co-pyrolysis of EN and PVC.

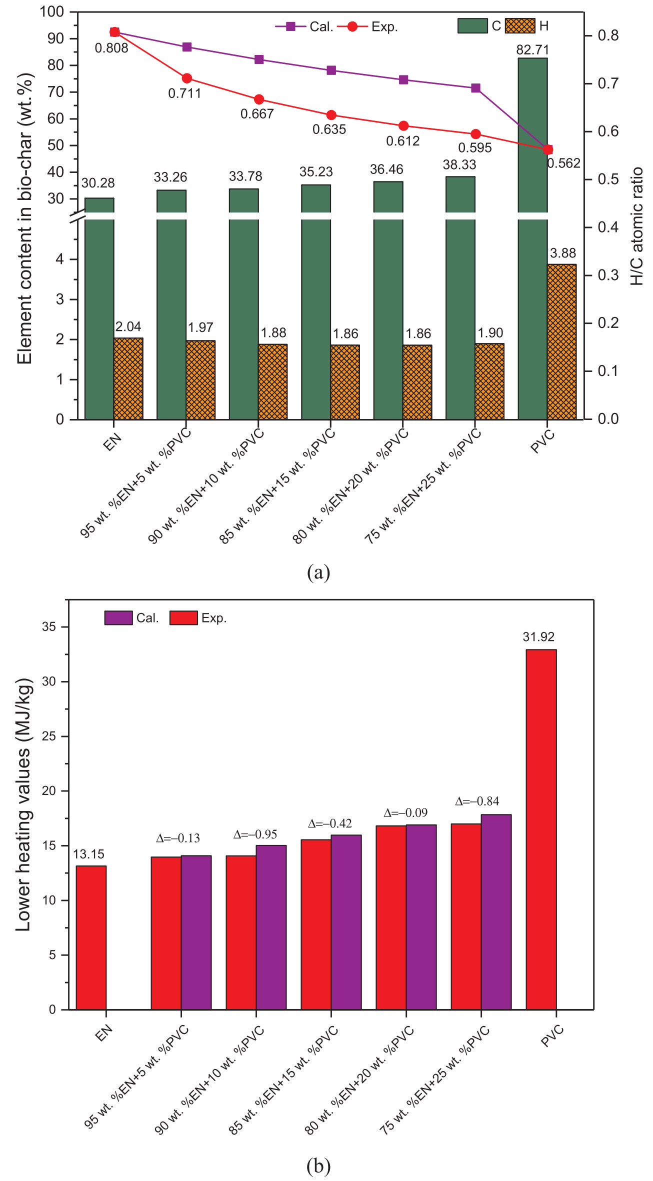

Fig. 8. Carbon and hydrogen contents, H/C atomic ratios (a) and lower heating values (b) of bio-chars obtained from co-pyrolysis of EN and PVC. (Δ = Exp.−Cal.)

nitrogenated compounds. Other oxygenated compounds contained organic acids, furans, ketones and aldehydes. The experimental values of relative contents of different groups existed in bio-oils from co-pyrolysis of EN and PVC were compared with the linear calculated values to investigate the synergistic effects, as shown in Fig. 7.

The relative content of aromatic hydrocarbons in EN bio-oil was only 3.44% and they were mainly light aromatics (1 ring). The relative content of aromatic hydrocarbons in PVC bio-oil was amazingly high up to 95.28%, which contained heterocyclic aromatics, light aromatics (1 ring) and light polycyclic aromatic hydrocarbons (PAHs) compounds (2–3 rings). For the bio-oils from co-pyrolysis, the relative contents of aromatic hydrocarbons increased from 12.76% to 25.43% as the content of PVC increased from 5 wt% to 25 wt%. The experimental values were higher than the calculated except the case of 25 wt% of PVC addition. Interestingly, there was no PAHs in bio-oils obtained from co-pyrolysis process. This is good for promoting the quality of bio-oils because the lesser concentration or absence of PAHs in bio-oils is eco-friendly and favorable due to more stability during combustion. The reason why the relative contents of light aromatics increased in bio-oils during co-pyrolysis process was that HCl evolved from PVC could accelerate the decomposition of EN and make heterocyclic or polycyclic aromatic to form light aromatics.

The relative contents of aliphatic hydrocarbons in EN and PVC bio-oils was only 1.42% and 4.72%, while aliphatic hydrocarbons in co-pyrolysis oils increased significantly, which ranged from 7.62% to 27.19%. The experimental values were higher than the calculated in all cases. When the addition content of PVC was 15 wt%, the relative content of aliphatic hydrocarbons was the highest (24.19%) and the difference between the experimental and the calculated was maximum. The increase of aliphatic hydrocarbons in co-pyrolysis might be due to the role of metal oxide in EN bio-chars, which could promote the formation of aliphatic hydrocarbons.

The relative contents of phenols in EN bio-oil was 19.35%, while there were no phenols in PVC oil. After PVC was added in the pyrolysis process, phenols increased by 10.23%–14.97% as the contents of PVC was increased from 5 wt% to 25 wt%. In addition, the experimental values were all higher than the calculated.

The relative content of other oxygenated compounds in EN bio-oil

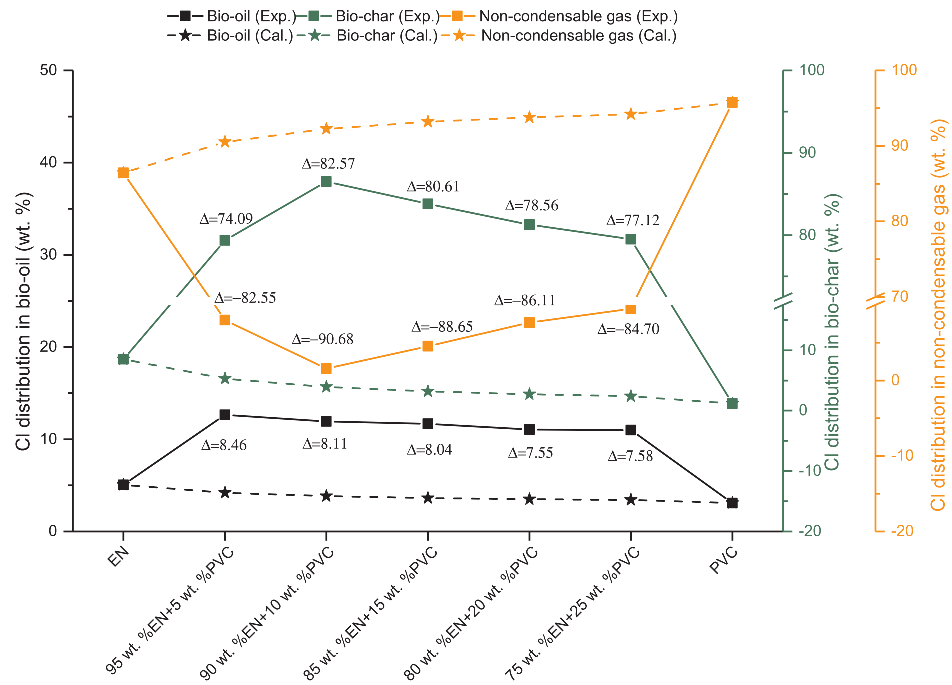

Fig. 9. Effect of the addition content of PVC on Cl distribution in bio-oil, bio-char and non-condensable gas. (Δ = Exp.−Cal.)

was the highest, which meant that the bio-oil obtained from EN alone had high content of oxygen. This resulted in the utilization limitation of bio-oils due to poor quality in terms of fuel properties. However, the bio-oils obtained from co-pyrolysis of EN and PVC was of lower contents of oxygenated compounds, and the experimental values were lower than the calculated. Amongst, the difference between the experimental and the calculated was minimum in the case of 15 wt% PVC addition. This was possibly due to synergistic effect of EN and PVC co-pyrolysis, which could promote the conversion of oxygenated compounds, such as organic acids, ketones and aldehydes, through accelerating C–O bonds breaking.

The relative content of nitrogenated compounds in EN bio-oil was up to 20.24%, while there were no nitrogenated compounds in PVC pyrolysis oil as PVC contained no nitrogen from its element analysis in Table 1. The experimental values were lower than the calculated, which means that co-pyrolysis of EN and PVC could reduce the nitrogenated compounds in bio-oils. The contents of nitrogenated compounds decreased to the lowest (4.59%) when the addition content of PVC was 25 wt%. This phenomenon was due to the interaction during the co-pyrolysis of EN and PVC. The reduction of nitrogenated compounds in bio-oils was favorable to the consequent utilization regarding to secondary pollution caused by nitrogen. In other research, the nitrogenated compounds in bio-oils during co-pyrolysis of protein and PVC were also reduced because of the existence of PVC which could suppress the formation of nitrogenated compounds [7].

It can be concluded from the above illustration that the synergistic effects during co-pyrolysis of EN and PVC promoted the formation of aromatic/aliphatic hydrocarbons and phenols, but suppress the oxygenated and nitrogenated compounds in bio-oils.

3.4. Characterization of bio-char from co-pyrolysis

The carbon/hydrogen elements and lower heating values of bio-chars obtained from the co-pyrolysis of EN and PVC were illustrated in Fig. 8(a), (b). The carbon and hydrogen contents in EN bio-char was 30.28 wt% and 2.04 wt%, while those in PVC char was higher, which were up to 82.71 wt% and 3.88 wt%, respectively. The contents of carbon in bio-chars obtained from the co-pyrolysis of EN and PVC increased while those of hydrogen were reduced slightly with the increase of PVC content. This phenomenon was also observed in research by Lu P et al. [22] on the co-pyrolysis of pine wood and PVC. The experimental and calculated values of H/C atomic ratio were also determined to investigate the synergistic effects of co-pyrolysis of EN and PVC on the properties of bio-chars. H/C atomic ratio of EN bio-char was 0.808, while that of PVC char was 0.562. Generally, a low H/C atomic ratio meant that the chat consisted predominantly of fixed carbon aromatic rings and thus was chemically stable [22,34]. The H/C atomic ratio of bio-chars decreased from 0.711 to 0.595 as the contents of PVC increased from 5 wt% to 25 wt%. In addition, the experimental values of H/C atomic ratio were lower than the calculated. The reason for this phenomenon might be that HCl evolved from PVC accelerated the dehydration of EN which diminished the hydrogen atoms of bio-chars [35]. The results indicated that the synergistic effect of co-pyrolysis of EN and PVC enhanced the aromatic degree of bio-chars.

Fig. 8(b) presents the lower heating values of bio-chars pyrolyzed from EN and PVC. The lower heating value of EN bio-char was 13.15 MJ/kg, while that of PVC char was high up to 31.92 MJ/kg. The lower heating values of bio-chars derived from co-pyrolysis of EN and PVC were increased with the addition increasing of PVC. The experimental of lower heating values were lower than the calculated, which was consistent with the results of H/C atomic ratio.

3.5. Cl content in three phase products

The distribution of chlorine in bio-oil, bio-char and non-condensable gas from the co-pyrolysis of EN and PVC when the addition of PVC increases from 5 wt% to 25 wt% which were presented in Fig. 9, and the experimental and calculated values were also compared to determine the synergistic effects on the Cl distribution. The contents of chlorine in bio-oils and bio-chars were detected using potentiometric titration after combustion. The Cl distributions in EN bio-oil, bio-char and non-condensable gas were 5.06 wt%, 8.50 wt% and 86.44 wt% whereas the Cl distributions in PVC bio-oil, bio-char and non-condensable gas were 3.08 wt%, 1.16 wt% and 95.76 wt%. This meant that most of Cl from EN or PVC was released in gaseous products during pyrolysis. The Cl distributions in three phase products changed significantly during co-pyrolysis of EN and PVC. When the addition of PVC was 5 wt%, 79.37 wt% of Cl from EN-PVC blends was transferred to bio-

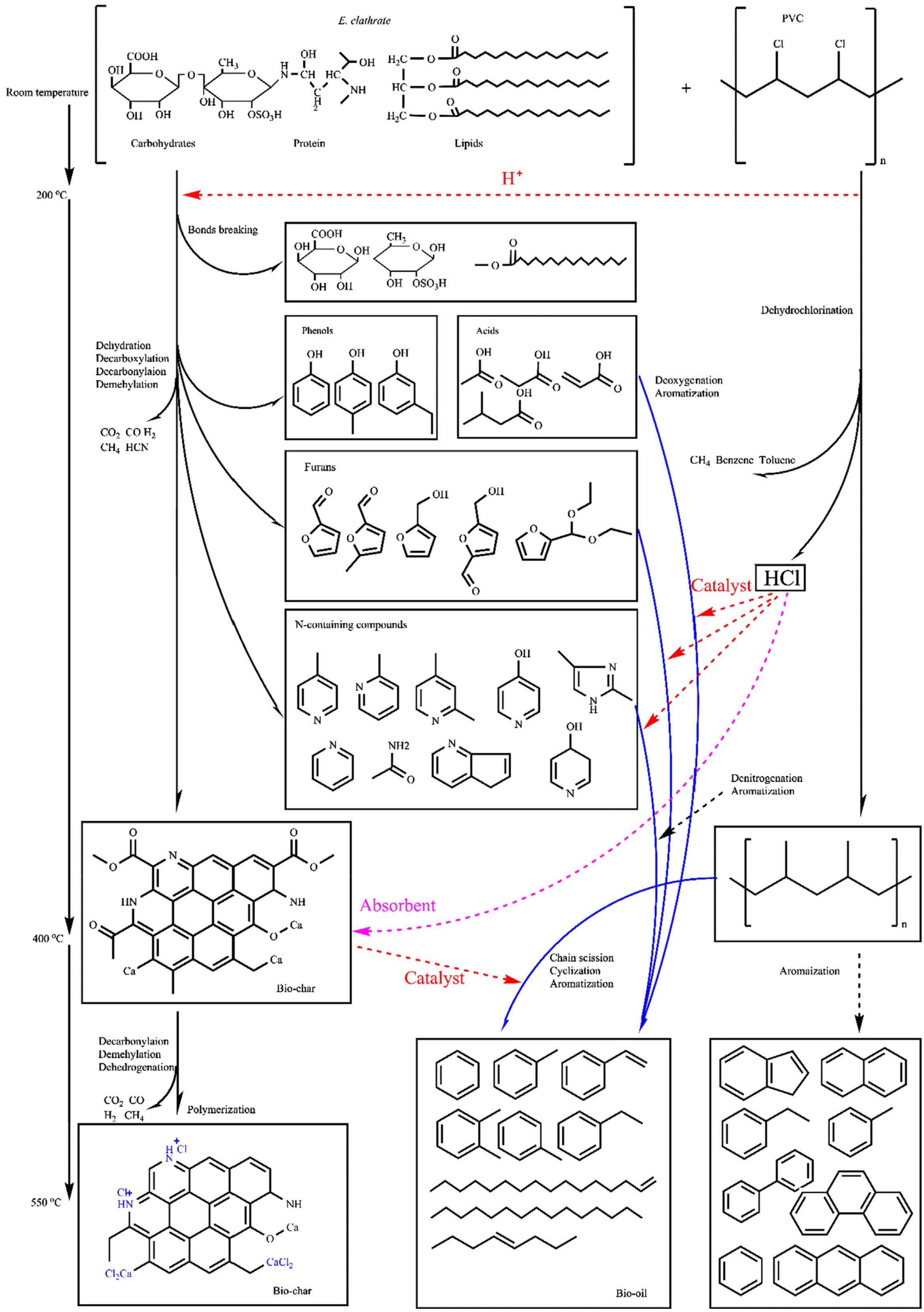

Fig. 10. Proposed reaction mechanism pathways during co-pyrolysis of EN and PVC.

char and 12.64 wt% to bio-oil, while the Cl distribution in non-condensable gas was 7.99 wt%. As the addition content of PVC increased to 10 wt%, the Cl distributions in bio-char increased to 86.50 wt% while that in non-condensable gas reduced to 1.57 wt%, which was the lowest in all experimental runs. The difference between the experimental and calculated values for bio-char and non-condensable gas in the case of 90 wt% EN + 10 wt% PVC was 82.57% and −90.68 wt%, which indicated that the synergistic effects were the most obvious in this case.

With the addition content of PVC increasing gradually, the differences between the experimental and calculated Cl distribution for bio-oil, bio-char and non-condensable gas decreased gently. The results indicated that the experimental Cl distributions in bio-oils and bio-chars were all higher than those through calculation, while the experimental Cl distribution in non-condensable gas was much lower than that through calculation. This meant that most of Cl in EN-PVC blends was transferred into bio-char and bio-oil due to the interaction between EN and PVC. Besides, the Cl distribution in bio-char was overwhelmingly higher than that in bio-oil. Hence, it is clear that most of Cl molecules from co-pyrolysis of EN and PVC were selectively trapped in the bio-char product. Our results indicated that Cl molecules could be largely fixed in bio-char product. The reasons could be described as follows; (1) During the release of HCl at low temperature, the residue obtained from EN in this stage could be as a strong Cl absorbent [36]; (2) As there existed an amount of metal compounds in bio-char from EN, Cl could be trapped in the inorganic solid phase, such as $CaCl_2$ [10]; (3) Previous researches indicated that there were some nitrogenated compounds in bio-chars from macroalgae, such as pyridinic-N, pyrrolic-N and quaternary-N [3]. Pyridinic-N is with strong coordination ability due to the lone electron pair in its structure, which could react with Cl^- producing to form the complex compounds.

The proposed reaction mechanism pathways during co-pyrolysis of EN and PVC were illustrated in Fig. 10. During co-pyrolysis of EN and PVC, HCl was released from PVC at low temperature stage ranging from 230 °C to 390 °C. This region was involved in the main thermal degradation of EN, when complex chemical reactions occurred, such as dehydration, decarboxylation and decarbonylation. HCl could serve as a kind of acid catalyst to suppress depolymerization and accelerate charring progress, resulting in lower H/C atomic ratio and heating values than the expected values. The bio-oils obtained from co-pyrolysis of EN and PVC in this period contained less oxygenated/nitrogenated compounds but more aromatic and aliphatic hydrocarbons due to the catalytic role of HCl. In addition, a majority of chlorine from EN-PVC blends was trapped in bio-char products, leaving less chlorine in bio-oil and gas products. This was favorable for the utilization of bio-oils and friendly to pyrolysis equipment and environment.

4. Conclusions

Synergistic effects on properties of bio-oil and bio-char during co-pyrolysis of EN and PVC were observed in DTG curves and pyrolysis experiments. In fixed bed experiments, interaction between EN and PVC resulted in less bio-oils but promoted more bio-chars due to HCl evolution from PVC. The synergistic effects could reduce the relative contents of oxygenated and nitrogenated compounds in bio-oils but promote those of phenols, aromatic and aliphatic compounds dramatically during co-pyrolysis of EN and PVC. Interaction during co-pyrolysis of EN and PVC enhanced the aromatic degree of bio-chars. Most of chlorine from EN-PVC blends was trapped in bio-char products. The synergistic effects during co-pyrolysis of EN and PVC revealed that co-pyrolysis technology is a potential approach to dispose residual macroalgae and Cl-containing plastics, which is a part of the major components of MSW.

Acknowledgement

This work was supported by National Natural Science Foundation of China (grant numbers 51676091).

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